



Student Performance Statistical Modeling

Comprehensive Study on Predictors of Exam Scores

Daniel D. Mondal

Roll No: 711

Executive Summary & Objectives

-  **Objective:** Develop high-accuracy predictive models for student “exam_score” using a large-scale dataset (N=80,000).
-  **Modeling Suite:** Comparative analysis of OLS, Stepwise Selection, Regularization (Ridge, LASSO, Elastic Net), and Non-Parametric (KNN) models.
-  **EDA Insights:** Identification of key drivers such as “previous_gpa” and behavioral factors (study hours, stress).
-  **Result:** Stepwise selection emerged as the optimal model with a Test RMSE of 4.1715.

Data Preparation & Methodology



Dataset Scope

N = 80,000 students

Cleaned records (no student_id)

Target: **exam_score**



Evaluation Split

80/20 Train-Test Partition

Train: 64,001 Rows

Test: 15,999 Rows

Standard seed (2026) ensured exact reproducibility across all modeling iterations.

Distribution of Final Exam Scores



Insight: The data shows a heavy left skew with a "Ceiling Effect" at 100. The mean score is approximately 89, indicating a high-performing population.

Key Correlation Drivers (r)

Variable	Correlation Type	Coefficient (r)
Previous GPA	Strong Positive	0.93
Motivation Level	Moderate Positive	0.25
Study Hours/Day	Moderate Positive	0.24
Exam Anxiety	Moderate Negative	-0.24
Stress Level	Slight Negative	-0.12

Bivariate Performance Trends



GPA Dominance

Near-perfect linear relationship explains why previous GPA is the primary predictive feature.

Study Habits

Clear positive slope confirms that parametric models (OLS, LASSO) are highly appropriate for this data.

Categorical Analysis: Major & Quality



Major Impact

Median scores remain consistent across Arts, Biology, Business, CS, Engineering, and Psychology.



Digital Infrastructure

Internet quality shows slight variation, necessitating dummy variable encoding for predictive accuracy.

Baseline Model: OLS & Multicollinearity

Variable	GVIF
Screen Time	23.0488
Study Hours/Day	13.1008
Social Media Hours	7.2637
Motivation Level	6.5411

4.174

OLS Baseline
RMSE

High VIF for Screen Time suggests redundancy
in numeric predictors.

Feature Reduction: Stepwise Selection

Backward AIC Method

Started with a full model and iteratively removed features that did not contribute significantly to minimizing the Akaike Information Criterion (AIC).

Benefit: Reduces model complexity without sacrificing predictive power, improving interpretability.

4.1715
Optimal Test RMSE

Regularization & Shrinkage Methods

Method	Mechanism	Test RMSE
Ridge (L2)	Shrinks coefficients towards zero to handle collinearity.	4.2901
LASSO (L1)	Feature selection by forcing irrelevant variables to zero.	4.1723
Elastic Net	Hybrid penalty combining L1 and L2 advantages.	4.1722

Elastic Net and LASSO significantly outperform Ridge, confirming that some predictors were redundant.

KNN: The Non-Parametric Approach

5%

Stratified Sample

CV

5-Fold Cross-Val

K=15

Optimal Tune

8.1976

KNN Test RMSE

Poorer performance suggests the data structure is better captured by linear models.

Final Model Performance Comparison

Model Rank	Model Technique	Test RMSE
1	Stepwise Selection	4.1715
2	Elastic Net	4.1722
3	LASSO (L1)	4.1723
4	OLS Baseline	4.1740
5	Ridge (L2)	4.2901
6	KNN (Sampled)	8.1976

Conclusion: Parametric reduction methods yield the highest predictive accuracy.